JavaScript Arrays

What is an Array?

An array is a special variable that can store multiple values in a single variable, instead of declaring separate variables for each value.

```
let fruits = ["apple", "banana", "mango"];
```

Here, fruits is an array that holds 3 strings.

Why Use Arrays?

- To group related data (like a list of items)
- Easily loop and perform actions on each element
- Makes code shorter, organized, and efficient

Array Creation

Using Square Brackets (Most Common)

```
let numbers = [10, 20, 30];
```

Using new Array() (Less Common)

```
let numbers = new Array(10, 20, 30);
```

Accessing Elements

Use index numbers to access individual items (index starts at 0).

```
let colors = ["red", "green", "blue"];
console.log(colors[0]); // red
```

```
console.log(colors[2]); // blue
```

Modifying Elements

You can change values using the index:

```
colors[1] = "yellow";
console.log(colors); // ["red", "yellow", "blue"]
```

Array Length

Returns the number of elements in the array.

```
let fruits = ["apple", "banana"];
console.log(fruits.length); // 2
```

Looping Through Arrays

Using for loop

Repeats a block of code a set number of times.

```
 for (let i = 0; i < fruits.length; i++) {
  console.log(fruits[i]);
}
```

Use when: You know how many times to loop (fixed count).

Using for...of

Loops through values of an array, string, or other iterable.

```
let colors = ["red", "green", "blue"];

for (let color of colors) {
  console.log(color);
}
```

Output:
red green blue

Use when: You want each value (not index) in an array or string.

Array Methods

push() – Add at end

```
let arr = [1, 2];
arr.push(3); // [1, 2, 3]
```

pop() – Remove from end

```
arr.pop(); // [1, 2]
```

unshift() – Add at start

```
arr.unshift(0); // [0, 1, 2]
```

shift() – Remove from start

```
arr.shift(); // [1, 2]
```

---

indexOf() – Find index of item

```
let items = ["a", "b", "c"];
console.log(items.indexOf("b")); // 1
```

includes() – Check if item exists

```
items.includes("c"); // true
```

---

slice(start, end) – Extracts part of array

Returns a new array without modifying the original.

```
let letters = ["a", "b", "c", "d"];
let part = letters.slice(1, 3); // ["b", "c"]
```

splice(start, deleteCount, ...items) – Add/Remove

Changes original array.

```
letters.splice(1, 2, "x", "y");
console.log(letters); // ["a", "x", "y", "d"]
```

---

concat() – Merge arrays

```
let a = [1, 2], b = [3, 4];
let c = a.concat(b); // [1, 2, 3, 4]
```

join() – Convert to string

The join() method is used to combine all elements of an array into a single string, using a separator (like comma, space, or dash).

```
let a = [1, 2];
let result = a.join("-");
console.log(result); // "1-2"
```

reverse() – Reverse order

```
a.reverse(); // [2, 1]
```

sort() – Sort items

The sort() method is used to rearrange (sort) the elements of an array in place. By default, it sorts alphabetically (as strings).

```
let numbers = [5, 8, 1, 4];

numbers.sort((a, b) => a - b); // [1, 4, 5, 8] (ascending)
numbers.sort((a, b) => b - a); // [8, 5, 4, 1] (descending)
```

Explanation of sort((a, b) => a - b)

It compares two values at a time (a and b)

If the result is:

- negative → keep a before b
- positive → swap them
- zero → keep order the same

So a - b sorts in ascending order
Use b - a for descending order

Advanced Array Methods

forEach() – Loop (no return)

The forEach() method runs a function once for each element in an array.
It's mainly used for looping through arrays.

```
let fruits = ["apple", "banana", "mango"];

fruits.forEach((fruit, index) => {
  console.log(index + ": " + fruit);
});
```

Output:
  0: apple

1: banana
2: mango

---

map() – Transform items

Returns a new array by applying a function to each item.

```
let nums = [1, 2, 3];
let doubled = nums.map(n => n * 2);
console.log(doubled); // [2, 4, 6]
```

---

filter() – Get items matching condition

The filter() method creates a new array that includes only the elements that pass a condition (return true).

```
let nums = [1, 2, 3];
let evens = nums.filter(n => n % 2 === 0);
console.log(evens); // [2]
```

---

reduce() – Single value from all items

```
let sum = nums.reduce((acc, curr) => acc + curr, 0); // 6
```

---

find() – Find first match

The find() method returns the first element in an array that matches a condition. If no match is found, it returns undefined.

```
let nums = [1, 2, 3];
```

```
let result = nums.find(n => n > 1);
console.log(result); // 2
```

Only one value is returned
Returns the first match only

ES6 Array Features

Destructuring

Assign array values to variables.

```
let [a, b] = [10, 20];
```

Spread Operator (...)

Copy or merge arrays.

```
let x = [1, 2];
let y = [...x, 3]; // [1, 2, 3]
```

Multidimensional Arrays

An array inside another array.

```
let matrix = [
  [1, 2],
  [3, 4]
];
console.log(matrix[1][0]); // 3
```

Summary Table of Common Methods

Method	Description
push()	Add to end
pop()	Remove from end
unshift()	Add to start
shift()	Remove from start
slice()	Extract portion (no change)
splice()	Add/Remove items (changes array)
map()	Transform each item
filter()	Filter items based on condition
reduce()	Reduce to single value
forEach()	Loop through each item
sort()	Sort items
concat()	Combine arrays
join()	Convert to string